MANAN SHETH

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PROFESSIONAL SUMMARY

AI/ML Engineer candidate (B.Tech CSE, FinTech & AI specialisation, GPA 8.9/10) who builds and rigorously validates deep-learning and Generative AI systems: PyTorch time-series models, walk-forward backtesting on NSE data, RAG pipelines and LLM-powered financial apps. IEEE-submitted researcher, ET Gen AI Hackathon national finalist and 2026 World Bank Youth Summit delegate. Seeking an AI/ML Engineer internship.

TECHNICAL SKILLS

AI / ML: PyTorch, LightGBM, Scikit-learn, TensorFlow, Temporal Convolutional Networks, LSTM, Transformers, Reinforcement Learning, Time-Series Forecasting, Generative AI, LLMs, RAG, LangChain, Ollama, FAISS, Embeddings, Recommendation Systems, Computer Vision
Languages: Python, SQL, TypeScript, Java, C++, C, Dart
Backend & Apps: FastAPI, Flask, Django, Node.js, React, React Native (Expo), Flutter, PySide6, Streamlit, Zustand, NumPy
Cloud & Tools: AWS, Google Cloud Platform, Docker, Git, PostgreSQL, MongoDB, SQLite, Firebase, Power BI, PyInstaller

PROJECTS

NiveshRL: Deep-Learning Research Terminal for Indian Equities Python, PyTorch, LightGBM, PySide6

- Built a Windows desktop research terminal (PySide6; packaged with PyInstaller and Inno Setup) that trains and monitors 24 models on NSE data, each tested walk-forward on point-in-time NIFTY 200 members after real transaction costs (paper trading only).
- Designed a dilated causal Temporal Convolutional Network (28k parameters, multi-task BCE + Huber heads, isotonic calibration) on triple-barrier labels for intraday trade selection; doubled LightGBM's information coefficient (0.032 vs 0.013) and moved loss per trade from −0.197R (random entries) to +0.016R after costs.
- Built an LSTM volatility forecaster that beat GARCH(1,1) by 16% on RMSE (0.337 vs 0.400) across 28,461 out-of-sample forecasts; its output drives stop-loss and position sizing.
- Trained FFNN, LSTM and Transformer stock rankers with significant predictive power (rank-IC t-stat 3.78); benchmarking against momentum exposed survivorship bias that had inflated returns from 19.3% to 39.8%.
- Built a decision layer with a neural-linear Thompson-sampling bandit, conformal abstention, implicit Q-learning exits and stacked meta-models; each is enabled only after passing a paired t-test, the rest run as shadow paper accounts.

FinGenie: AI Financial Literacy & Expense Monitoring App React Native (Expo), FastAPI, LangChain, Gemini

- Built a cross-platform app with a FastAPI backend and Zustand state; integrated Gemini and LangChain for a RAG-based financial news feed with text-to-speech and a memory-persistent, event-driven AI financial coach.
- Developed a NumPy engine for 30-year FIRE projections and a Portfolio X-Ray (XIRR, expense ratio via external APIs), plus a privacy-first offline SQLite expense tracker.

Tax Consultancy Application: Local RAG for Legal Documents Ollama, LangChain, FAISS

- Built a fully local RAG pipeline to parse complex tax legislation, returning source-verified, clause-level answers with 100% data privacy.

EDUCATION

GLS University, Ahmedabad, India 2023 – Expected 2027
B.Tech, Computer Science Engineering | GPA: 8.9/10

Surya Varsani Academy, Bhuj, India 2021 – 2023
*AS & A Levels (HSC equivalent), Grade A; IGCSE (SSC equivalent), 2021, Grade A**

RESEARCH & ACHIEVEMENTS

- **IEEE Paper:** Authored a study on the impact of Generative AI on Indian financial markets (under review).
- **ET Gen AI Hackathon, National Finalist (Round 2):** Developed FinMentor, a Generative AI financial literacy tool; one of the top shortlisted teams nationally.
- **2026 World Bank Youth Summit Delegate:** Contributed to policy discussions on “Future Works: Designing Jobs for the Digital Age”.
- **District-Level Gold Medalist, Swimming:** Represented Kachchh District at State Level.

CERTIFICATIONS

- AWS Certified Cloud Practitioner; Supervised Machine Learning (DeepLearning.AI); Machine Learning for Trading (Google); Google AI Essentials; Google UX Design; Power BI Data Analytics

LEADERSHIP & WRITING

- **Volunteer Coordinator, Parasdham Youth NGO:** Ran end-to-end logistics for community outreach and weekly volunteer drives.
- **Technical & Financial Content Writer:** Blog on Generative AI in financial markets (mananshethblogs.wordpress.com).